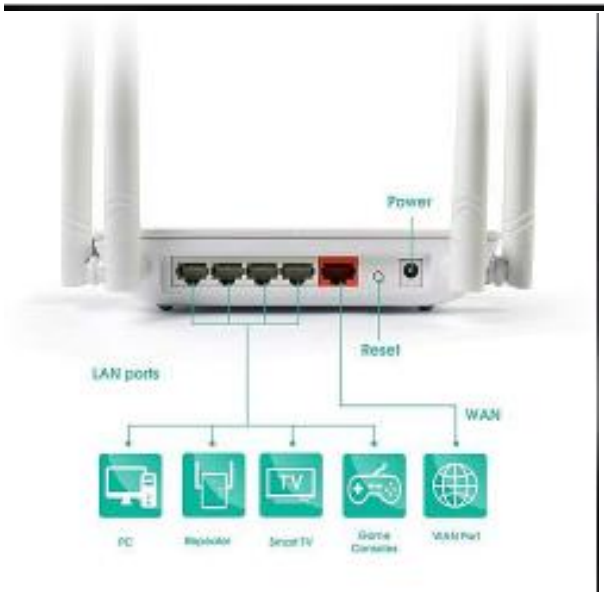
Day 2

Routers and SwitchesTheory (30 min):Basic functions and differences between routers and switches .Practical (30 min):Identify ports and LEDs on a router and switch.

TASK - 1

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.
➔





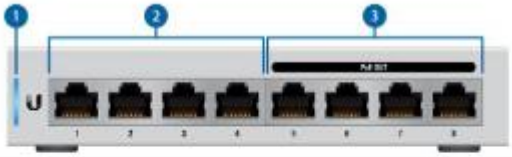


LED indicators

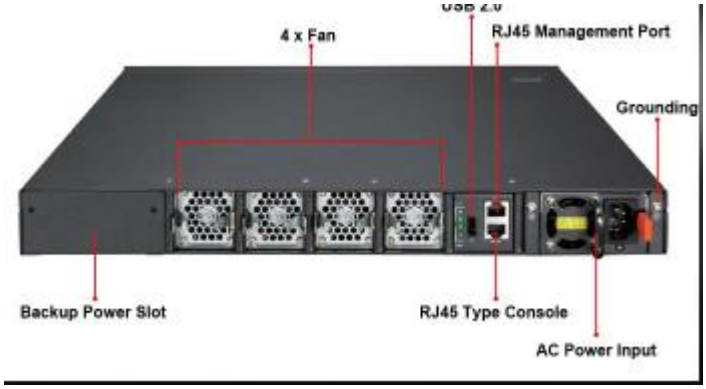


Gigabit ports

Front Panel



1 System LED	
Click here for detailed descriptions.	
2 RJ45 (Ports 1 - 4)	
RJ45 ports support 10/100/1000 Ethernet connections	
3 RJ45 (Ports 5 - 8)	
RJ45 ports support 10/100/1000 Ethernet connections and auto-sensing 802.3af PoE	



- Ethernet ports: Connect PCs, printers, routers, and other network devices.
- Port LEDs: Indicate link status and network activity for individual ports.
- Power port: Supplies electricity to the switch.
- Console port (managed switches only, if present): Used for command-line configuration and troubleshooting.
- Uplink port (if specifically marked): Used to connect the switch to another switch or network device. On many modern switches, any suitable Ethernet port can serve this role.

TASK -2

List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

➔

Feature	**Router**	**Switch**
Main Purpose	Connects Different Networks	Connect devices with a LAN
Data Handling	Uses IP to route Packets	A layer 2 switch uses a MAC address to transfer FRAMES
Typical Location	Between HOME, Office	HOME, OFFICE and Computer lab LAN
Example	Home wifi Router	8 port Ethernet Switch

TASK – 3

Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

➔

Component	**Location**	**Purpose**
WAN Port	Router Real Panel	Connect to modem or ISP
LAN Port	Router Real Panel	Connect wired devices or a switch
Power Port	Real panel of either device	Supplies electrical power
Status LED	Router front panel	Shows Power, Internet, Wifi, LAN Status
Ethernet Ports	Switch rear Panel	Connect computer and other wired devices
Link/ activity LEDs	Near Switch Ports	Show whether device is connected and where data is passing
Console ports	Some routers and managed Switches	Used to configure and troubleshoot device

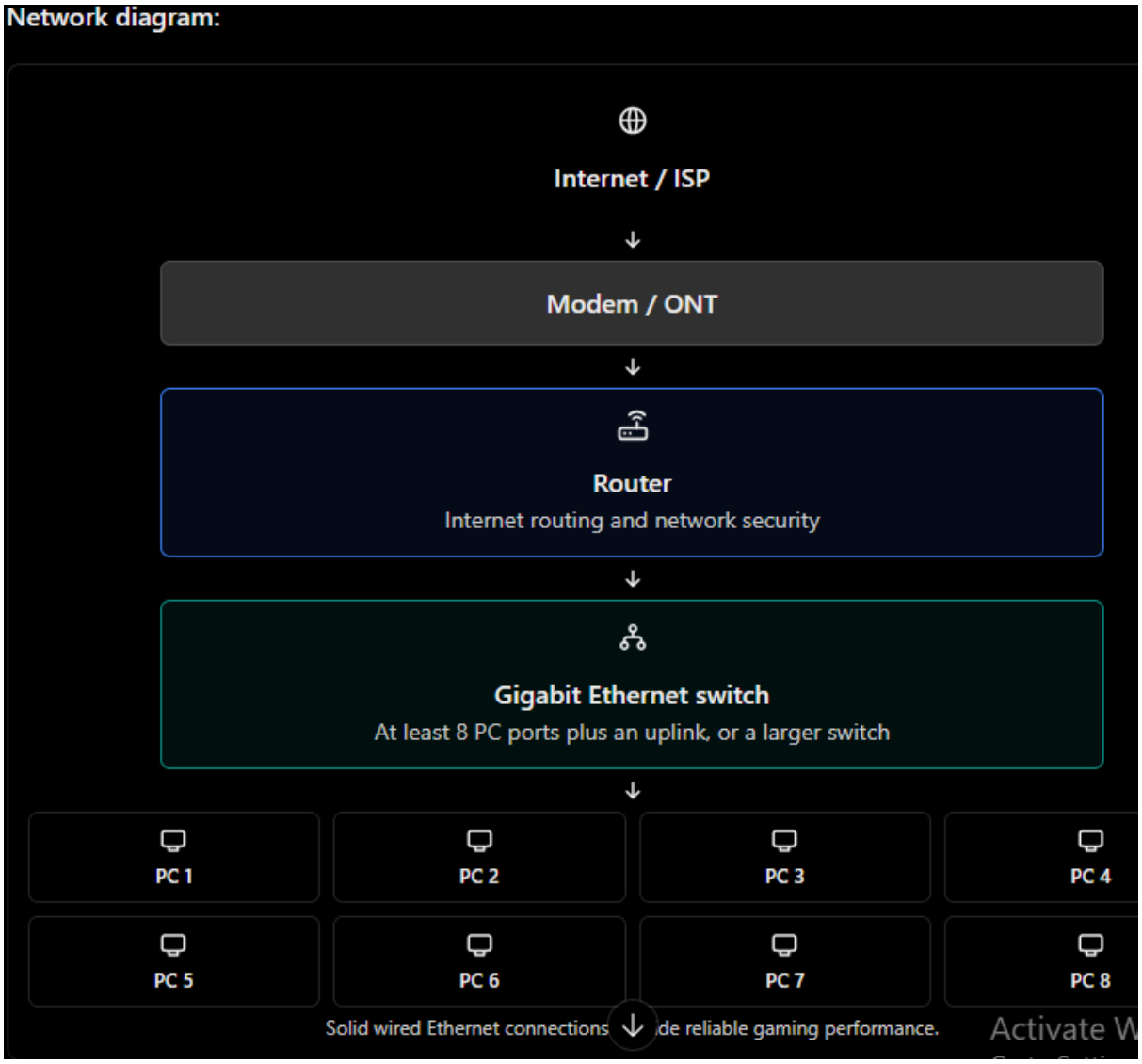
Ports allow devices and cables to connect, while LEDs help us check whether the device is powered on and whether network connections are working.

TASK -4

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

➔

I would connect the ISP internet line to the modem or ONT, then connect the modem or ONT to the router's WAN port. Next, I would connect one LAN port of the router to an Ethernet switch using a network cable. Finally, I would connect all eight gaming PCs to the switch using Ethernet cables.



1. A switch provides enough Ethernet ports to connect all eight gaming PCs.
2. Wired Ethernet generally offers more stable connections and lower latency than Wi-Fi.
3. The router manages internet access and routes traffic between networks.
4. A suitable gigabit switch supports fast connections between the PCs and router.

For a real gaming café, I would also configure appropriate security settings and consider traffic prioritization (QoS) if the router supports it. The internet plan must have enough bandwidth for gaming, streaming, and other users.